



UTM
UNIVERSITI TEKNOLOGI MALAYSIA

SECP2753
DATA MINING
SECTION 01

ASSIGNMENT 1

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A Case Study on Mushroom and CPU Performance Datasets

1.0 Introduction

In today's digital world, we collect a lot of data from different sources every day. This raw data, however, is often messy, unorganised, and hard to use directly. This is where data mining becomes important. Data mining is a powerful method used to find patterns, trends, or useful information hidden inside large datasets. It helps researchers, companies, and organizations make better decisions by learning from the data.

Before we can start analysing data or using it to build machine learning models, we need to prepare the data properly. This process is called data pre-processing. It involves cleaning the data, transforming it into the right format, and making sure it is consistent and accurate. Without proper pre-processing, the results of data mining can be misleading or incorrect.

In this assignment, we focus on two real-world datasets taken from the UCI Machine Learning Repository, which is a well-known source for open datasets used in machine learning and data analysis. The two datasets we have selected are:

1. Mushroom Dataset – This dataset includes different characteristics of mushrooms and indicates whether they are edible or poisonous. It can help us understand how different features relate to the safety of mushrooms.
2. CPU Performance Dataset – This dataset contains information about the performance of different computer processors based on technical features such as memory size, machine cycle time, and other related attributes.

First, we will explore both datasets to discover interesting patterns or insights. For example, in the mushroom dataset, we might find which features are most helpful in identifying poisonous mushrooms. In the CPU dataset, we might discover what features have the biggest effect on computer performance. Second, we will demonstrate data pre-processing techniques, focusing on two key methods: Normalisation and discretisation. These techniques help to prepare the data for building effective data mining models. For this part, we will use the CPU Performance dataset to give clear examples of how Normalisation and discretisation are applied and why they are important.

2.0 Dataset Overview and Insights

2.1 Mushroom Dataset

The Mushroom dataset contains information about 8,124 different mushrooms. Each mushroom is described by a set of 22 categorical features (also called attributes), and one target attribute that tells us whether the mushroom is edible (safe to eat) or poisonous.

All the features in this dataset are qualitative, which means they describe characteristics using categories (like shapes or colours) rather than numbers. Because of this, the dataset is well-suited for classification tasks, where the goal is to predict a category, such as edible vs. poisonous.

Some examples of the features include:

- Cap shape: This tells us the shape of the mushroom cap. It can be one of the following types: bell, conical, convex, flat, knobbed, or sunken.
- Odour: This describes the smell of the mushroom. Possible odours include almond, anise, creosote, fishy, foul, musty, none, pungent, and spicy.

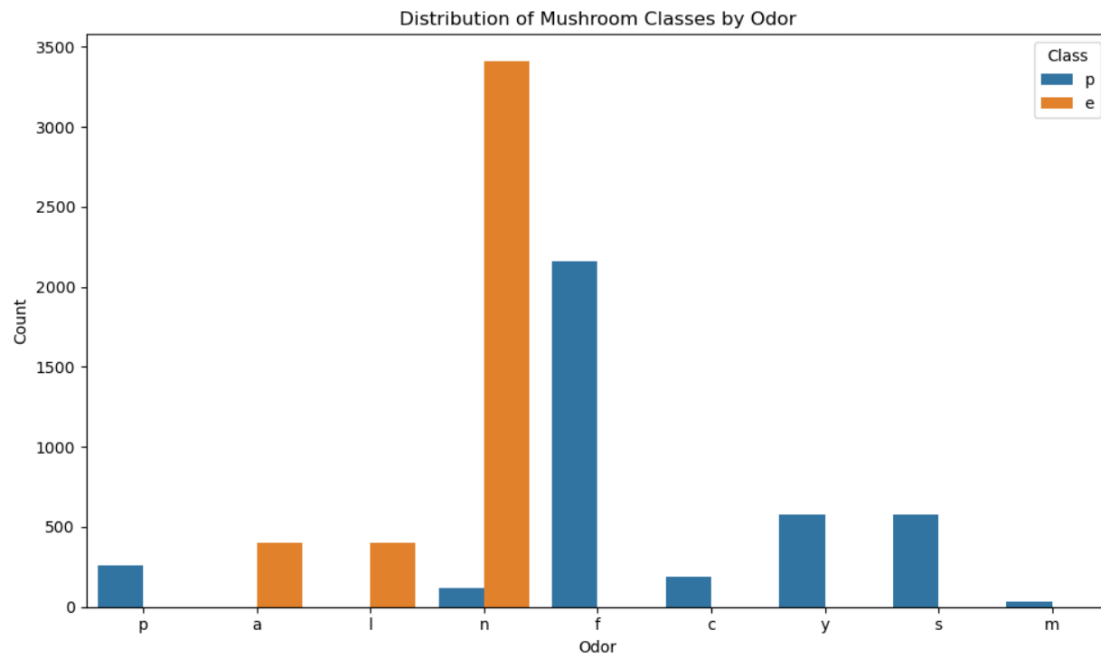
After exploring the dataset, we found some very useful insights. The 'odour' attribute is strongly related to whether a mushroom is poisonous or not. For example, mushrooms with a foul or creosote odour are almost always poisonous, while mushrooms with a pleasant smell like almond or no smell are usually edible.

Other features such as 'spore-print-color' (the colour left by the mushroom spores) and 'gill-size' (the size of the gills under the cap) also play an important role in predicting edibility. These features show clear patterns that help us separate edible mushrooms from poisonous ones.

Class Labels

Edible = e, poisonous = p

Bar Plot: Odour vs Class



Bar Plot: Gill-Size vs Class



2.2 CPU Performance Dataset

The CPU Performance dataset contains information about 209 different computer processors (CPUs). Each CPU is described using numerical features related to its hardware specifications and performance. The main goal of this dataset is to understand how these features affect the CPU's performance, which is given as the Published Relative Performance (PRP) score. Because the target variable (PRP) is numerical, this dataset is suitable for both regression tasks (predicting a number) and classification tasks (predicting performance levels, high or low).

Key Attributes to be analysed:

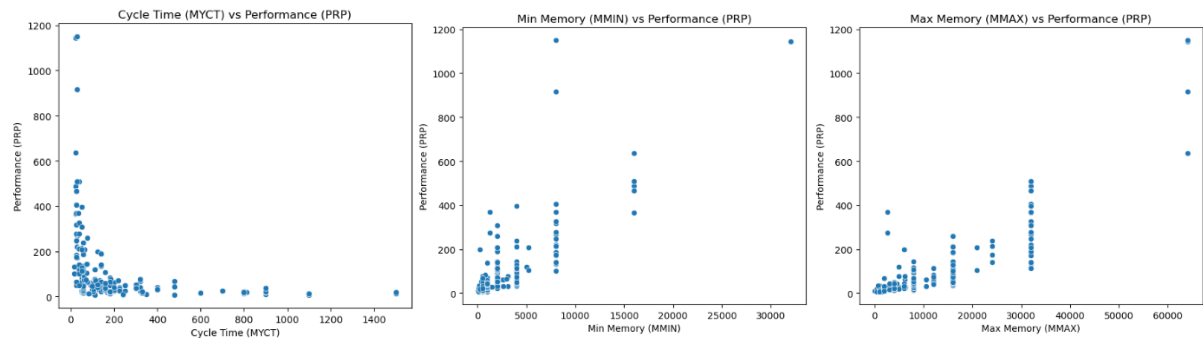
- **MYCT**: Machine cycle time (measured in nanoseconds). Lower values mean faster processing.
- **MMIN / MMAX**: Minimum and maximum main memory (in kilobytes). These represent the memory range available for a CPU.
- **CACH**: Cache memory size (in kilobytes). Larger cache sizes can improve performance.
- **CHMIN / CHMAX**: Minimum and maximum channel capacity. These values relate to the CPU's communication ability with other components.
- **PRP**: Published relative performance score. This is the **target variable** we want to predict.

From our data exploration, we discovered several important patterns. There is a strong inverse relationship between MYCT (cycle time) and PRP (performance score). This means that CPUs with lower cycle times (faster) tend to have higher performance scores.

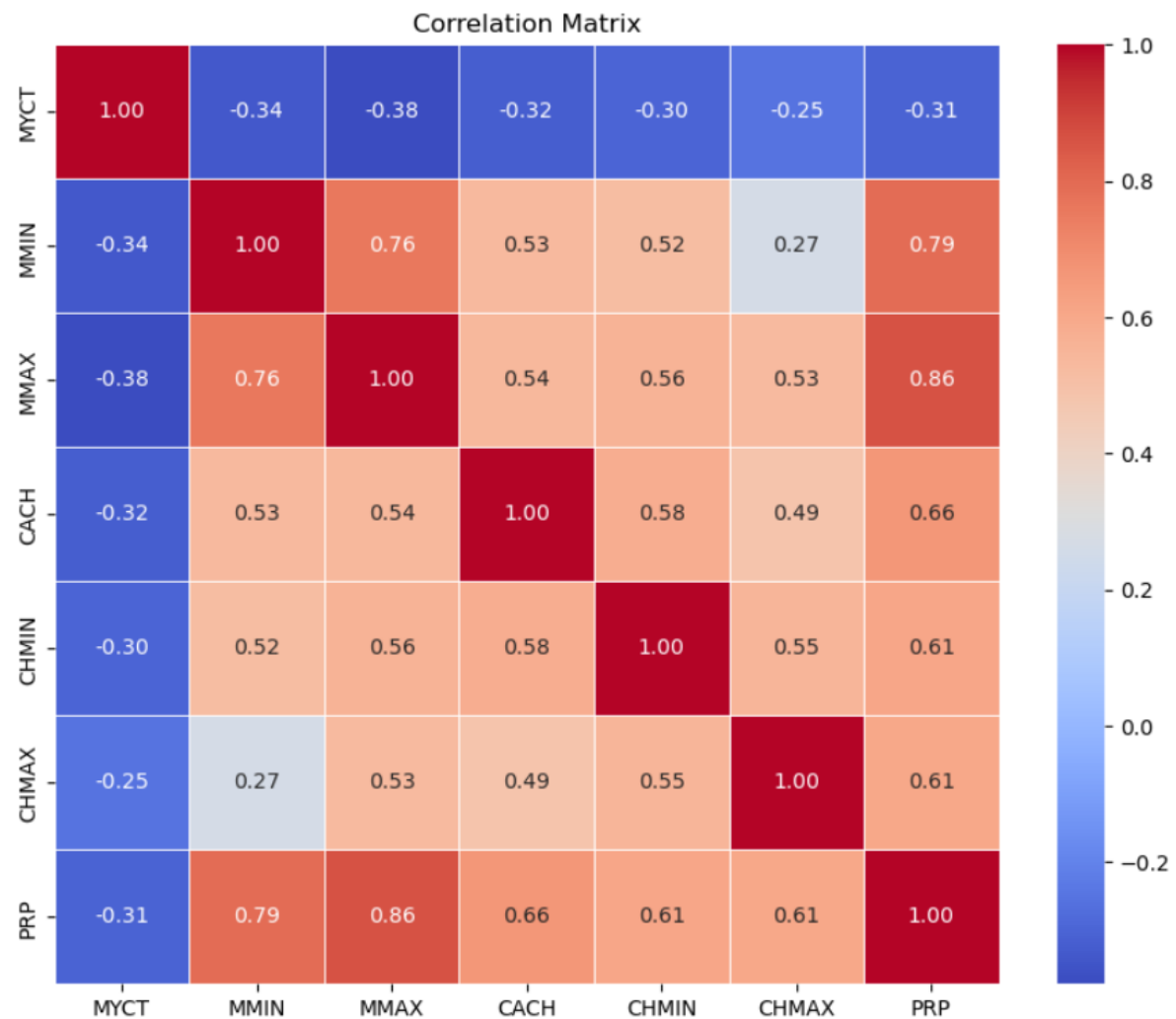
The memory features, MMIN and MMAX, show a positive correlation with performance. CPUs with larger memory tend to perform better, which shows the importance of memory size in overall computing power.

By using scatter plots and correlation matrices, we were able to visually identify which features have the strongest influence on the target variable. These tools help us understand which CPU specifications are most important for performance.

Scatter Plot



Correlation Matrices



3.0 Pre-processing Dataset: CPU Performance

Out of the two datasets we explored, we chose the CPU Performance dataset for detailed pre-processing. The main reason is that this dataset contains continuous numerical data, such as cycle time, memory size, and performance scores. These types of data are especially suitable for pre-processing techniques like normalisation and discretisation.

Normalisation helps scale the data into a standard range, usually between 0 and 1. This is important because it prevents features with large values (like memory size) from dominating features with smaller values (like cycle time) in the model.

Discretisation involves converting continuous data into categories or intervals. This can simplify the data and help some machine learning algorithms work better, especially if we plan to apply classification later on.

4.0 Pre-processing Techniques

4.1 Normalisation

Normalisation is a data pre-processing technique that changes numeric values into a standard scale, usually between 0 and 1. This is especially useful when working with datasets that have features measured in different units or ranges. Without normalisation, features with large values could dominate those with smaller values, which might affect the results of machine learning models.

For the CPU Performance dataset, we use the Min-Max Normalisation method. This technique adjusts the values of a feature so that the smallest value becomes 0, the largest becomes 1, and all other values are scaled proportionally between them.

$$\text{Normalised Value, } v' = \frac{v - \min_A}{\max_A - \min_A} (\text{new_max}_A - \text{new_min}_A) + \text{new_min}_A$$

where

v = original value,

$\min_A$ = Minimum value in attribute A,

$\max_A$ = Maximum value in attribute A,

new_max_A = New minimum value after normalisation,

new_min_A = New maximum value after normalisation

For example, we want to normalise MYCT (Machine Cycle Time) to the range [0,1].

Minimum value, $min_A = 17$

Maximum value, $max_A = 150$

New range = 0 to 1

Original value, $v = 75$

$$\text{Normalised Value, } v' = \frac{v - min_A}{max_A - min_A} (new_max_A - new_min_A) + new_min_A$$

$$\text{Normalised Value, } v' = \frac{75 - 17}{150 - 17} (1 - 0) + 0 = \frac{58}{133} \approx 0.436$$

$\therefore$ After normalisation, the MYCT value of 75 becomes approximately 0.436.

4.2 Discretisation

Discretisation is a pre-processing technique that transforms continuous numerical data into categorical data by grouping values into intervals or "bins." This method is useful for simplifying data, making it easier to understand, and it is especially helpful for models that work better with categories, such as decision trees or rule-based classifiers.

In this report, we applied Discretisation to the PRP (Published Relative Performance) attribute in the CPU Performance dataset. Instead of using the exact numerical values for PRP, we grouped them into three performance levels based on their range. Discretising PRP allows us to simplify the target variable by converting it from a wide range of numbers into just three clear categories.

- Low Performance: PRP value between 20 and 80
- Medium Performance: PRP value between 81 and 140
- High Performance: PRP value between 141 and 250

For example, if a CPU has a PRP score of 160, it falls into the High Performance category because 160 is within the range 141–250.

Results

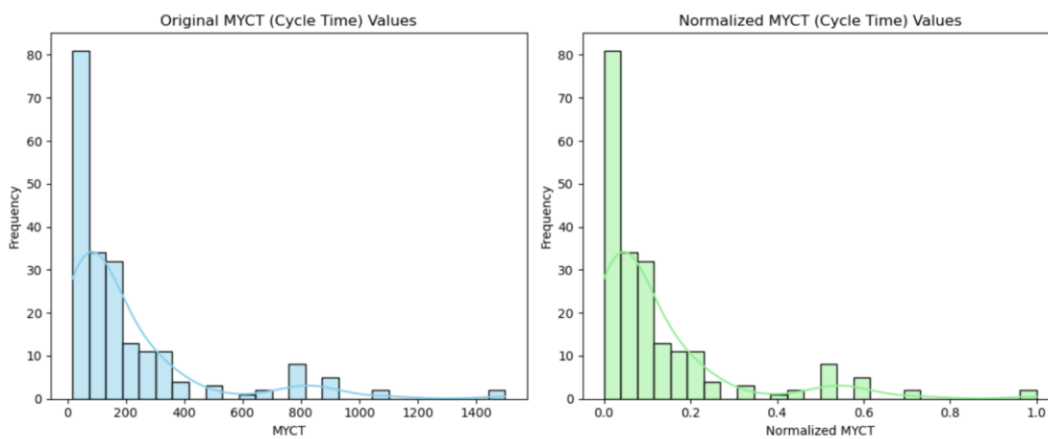
Normalized MYCT value for 75: 0.04

Discretized PRP value for 160: High

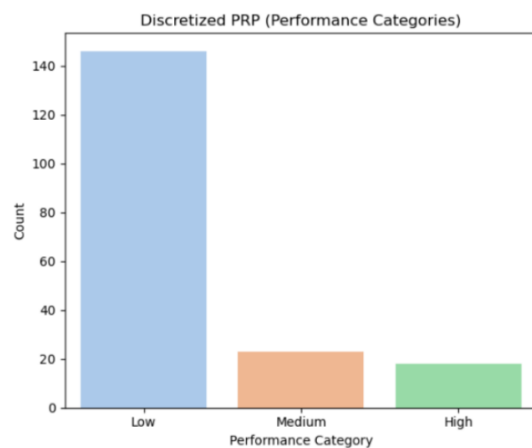
	VENDOR_NAME	MODEL_NAME	MYCT	MMIN	MMAX	CACH	CHMIN	CHMAX	PRP
1	adviser	32/60	125.0	256.0	6000.0	256.0	16.0	128.0	198.0
2	amdahl	470v/7	29.0	8000.0	32000.0	32.0	8.0	32.0	269.0
3	amdahl	470v/7a	29.0	8000.0	32000.0	32.0	8.0	32.0	220.0
4	amdahl	470v/7b	29.0	8000.0	32000.0	32.0	8.0	32.0	172.0
5	amdahl	470v/7c	29.0	8000.0	16000.0	32.0	8.0	16.0	132.0

	ERP	MYCT_normalized	PRP_discretized
1	199.0	0.072825	High
2	253.0	0.008092	NaN
3	253.0	0.008092	High
4	253.0	0.008092	High
5	132.0	0.008092	Medium

Normalisation graph



Discretisation graph



5.0 Discussion

Data pre-processing is an important step in preparing data for mining tasks. Normalisation is used to adjust the scale of features, which is important for algorithms that depend on distances, such as KNN and clustering. Discretisation, on the other hand, helps convert continuous data into easy-to-understand categories.

The CPU Performance dataset showed clear connections between hardware specifications and performance. Features like MYCT (machine cycle time) and memory size are important for predicting PRP (performance rating). Properly scaling and categorising these features can improve the model's performance and accuracy.

In contrast, the Mushroom dataset didn't require pre-processing but showed clear patterns in categorical data, which were helpful for classification. This highlighted the importance of understanding the relevance of each attribute before applying any algorithm.

6.0 Conclusion

This report explored two datasets to understand their features and the effect of pre-processing. The Mushroom dataset, which contains categorical data, was useful for learning how to build classification models using grouped information. The CPU Performance dataset highlighted the importance of pre-processing techniques such as normalisation and discretisation.

Normalisation scales the data to a consistent range, which helps models that rely on measuring distances between values. Discretisation converts continuous data into categories. It helps make the data easier to interpret and use in classification. These techniques improve model accuracy and make data easier to work with before analysis.

7.0 References

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- [4] Kaggle. (n.d.). Public datasets repository. <https://www.kaggle.com/datasets>